

Bioclean Smart Biogas Systems

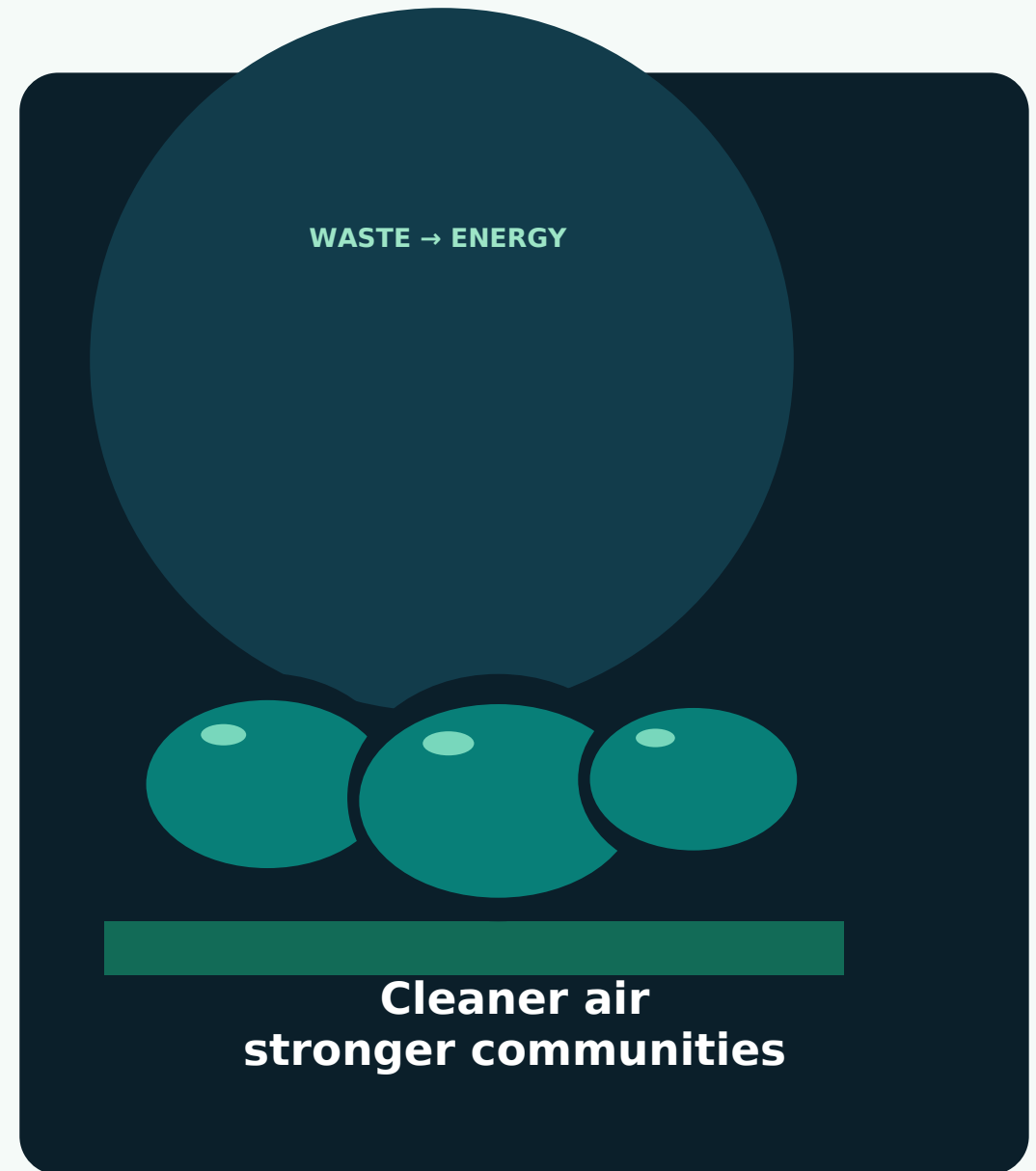
Combating air pollution in schools and communities in Enugu State

Creating a healthy environment for better learning and healthier living.

Presented by

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Why air pollution matters

Pollution affects health, learning, household costs and the balance of nature.

6.7M

premature deaths annually
cited in the supplied draft

Air pollution is the
release of substances
into the atmosphere in
quantities that can
harm human health,
modify the climate and
disturb nature.

01 PM2.5 / PM10

Tiny particles from vehicles and generators; can reach the lungs.

02 CO

Carbon monoxide from incomplete burning of coal, firewood, diesel and petrol.

03 SO₂

Sulphur dioxide from gas flaring and diesel.

04 NO₂

Nitrogen dioxide from vehicle engines, generators and gas-flaring industries.

05 O₃

Ground-level ozone, a major part of smog formed when sunlight reacts with pollutants.

06 CH₄

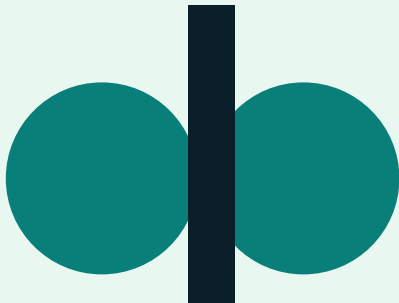
Methane from sewage, decomposing waste and gas flaring; contributes to ground-level ozone.

Current situation in my school

What students experience around hostels, open dumps and classrooms.

A learning environment under pressure

Clean air is not a luxury. It is a learning condition.



NH₃ · H₂S · smoke

Students report:

- Drowsiness and nausea, especially within the hostels.
- Possible exposure to ammonia (NH₃) and hydrogen sulphide (H₂S) from sewage and open dumps.
- Discomfort when smoke from burning waste is inhaled.
- Some students avoid classrooms because they cannot breathe comfortably.

The challenge is both environmental and educational.

My community challenge

Household energy choices and weak sanitation systems can compound air pollution.

01

Household energy

Many homes use charcoal, stoves and generators for cooking and powering their houses.

02

Carbon monoxide risk

Incomplete combustion in limited oxygen can release CO. Inhaled CO can form carboxyhaemoglobin and reduce oxygen delivery.

03

Sanitation challenge

Broken septic tanks and leaking sewage pipes are reported in many homes, contributing to pollution.

Community need → safer waste handling + cleaner energy + better sanitation

How air pollution affects schools and communities

The effects reach beyond the atmosphere.

01

Health effects

Asthma, pneumonia and lung cancer are among the health concerns associated with polluted air.

02

Academic effects

Polluted environments can make learning difficult, contribute to poor grades and reduce educational outcomes.

03

Economic effects

Families may spend income on treatment, while illness can reduce workers' productive days.

04

Social effects

Smoke, smog and odour can reduce interaction and contribute to fear, embarrassment and stigma.

Research behind the innovation

BSBS turns organic waste into a clean-energy pathway.

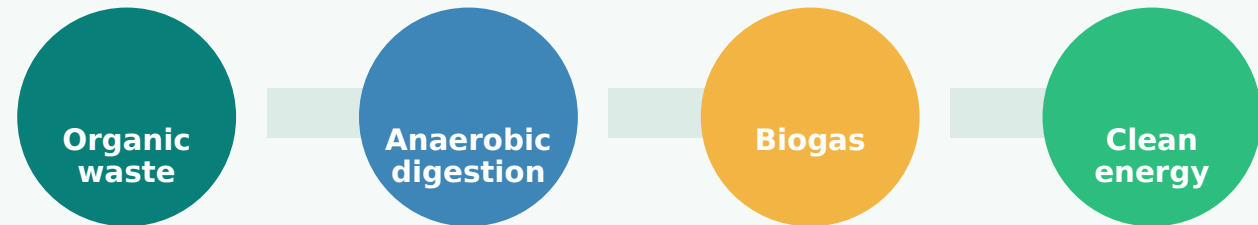
BSBS

Bioclean Smart Biogas System

Biogas is a renewable fuel produced when microorganisms break down waste without oxygen.

A local waste problem becomes a local energy resource.

THE PROCESS



Without oxygen

Microorganisms transform biodegradable material into a usable fuel and a nutrient-rich residue.

Key metrics & sustainability plan

The handwritten notes point to shared ownership, training and transparent monitoring.

01 Student leadership

Organise a reliable student team and Green Club to help manage operations.

02 Community training

Train community members in safe operation, maintenance and installation.

03 Local partnerships

Engage schools, households, farmers and community partners.

Suggested measures to track

Waste diverted Organic material kept out of open dumps

Clean energy produced Biogas generated for schools and homes

Savings & revenue Financial performance tracked monthly

People trained Operators and installers supported locally

Financial viability & sustainability

Draft figures retained from the handwritten notes; reconcile the calculation before formal submission.

**Pays for itself
in 12 months**

Investment case

Savings + revenue can support
cleaner energy, community

Transparent, app-based
monitoring

₦1,500,000

Savings

No waste truck / collection savings

₦320,000

Revenue

Organic fertiliser + carbon credits

₦50,000

Running cost

Maintenance and operating costs

₦1,580,000

Reported net gain / year

Figure transcribed from notes

Note: the reported figures do not currently reconcile arithmetically; confirm the final model before presenting.

How BSBS combats air pollution

A clear before-and-after pathway from waste to cleaner living.

BEFORE

Waste burning → Smoke → Air pollution → Health problems

AFTER

Waste collection → Biogas → Clean energy → Clean environment

Benefits of the BSBS model

- Health: less smoke from firewood, charcoal and generators.
- Access: intended to be affordable for families and schools.
- Reported net benefit: ₦1,850,000 per model in a home or community.
- Jobs: creates opportunities for trained local installers.
- Environment: slurry can be dried and sold to farmers.
- Accountability: app-based monitoring supports transparency.

Cleaner schools. Healthier communities.

Bioclean Smart Biogas Systems can connect waste management, clean energy, better learning conditions and local opportunity.

Thank you

- 01 **Confirm** Reconcile the financial model and final figures.
- 02 **Pilot** Test the system in a school or community setting.
- 03 **Train** Equip the Green Club and local operators.
- 04 **Measure** Track air quality, savings, energy and slurry use.